

[]

ATE-SS Asent

✓ []

At. Scaled

LLM

→ NN

→ Gen NN - nano gpt

→ LLM - GPT 2

✓ Pretrain (1)
✓ Instruction Tuning (2)
Preference Tuning (3)

SI

sh
den
orm
ans
gema

1.75

Smolm2

→ (Fine-tuning) LLM

8k
an
1k
scale

Domain Customization

dt.

✓ rewards

habits

Scum

(Int. Calls)

Sel say

Calc-s

Lora
9 core

Conversation
+
Int. Calls

from: system

rule: —

<rule>

[& type: "rule"

name: —

sketch

id: —

param: —

{

rule

{

Cancel

{

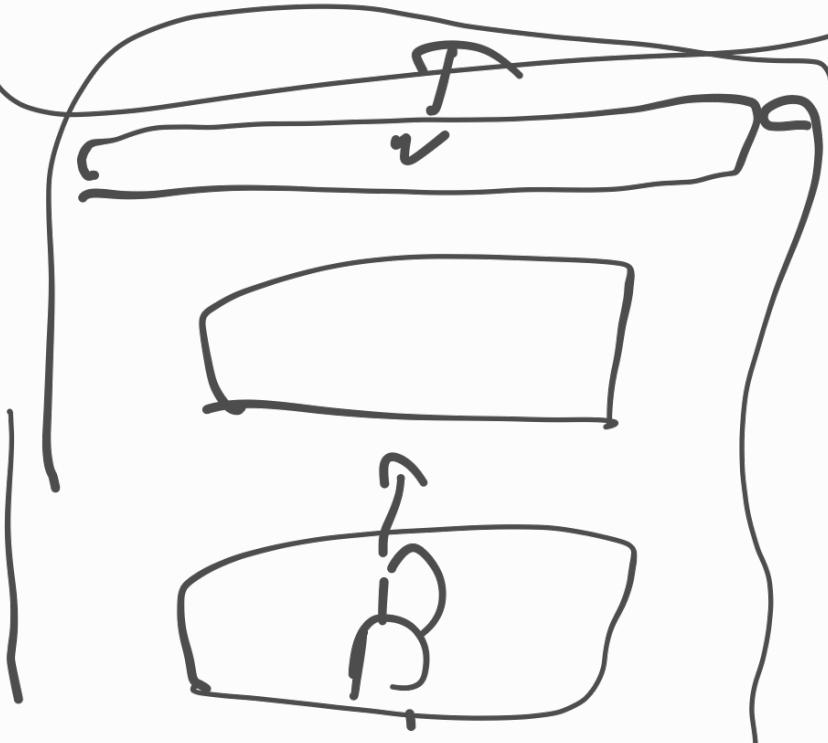
from: canvas.sch

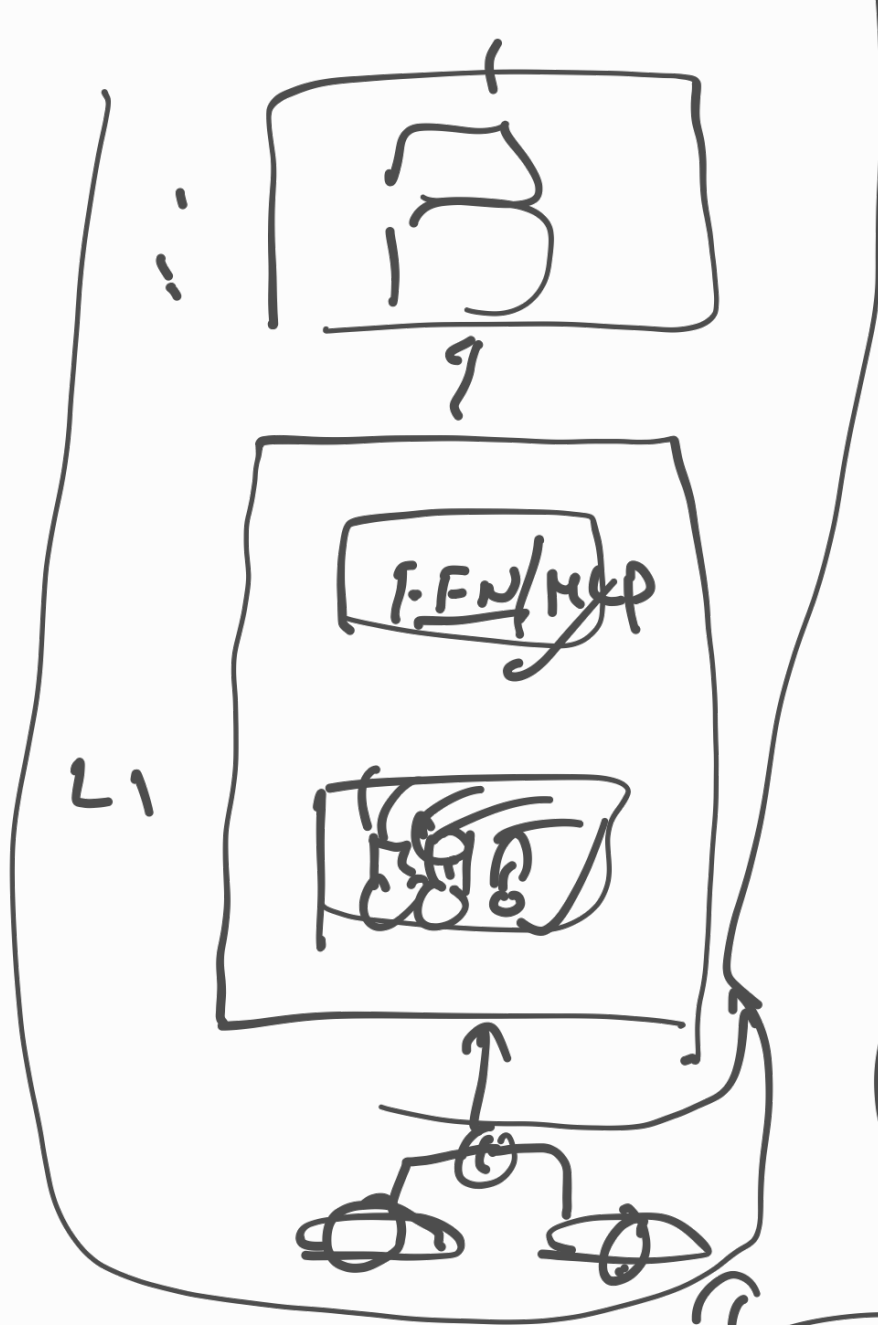
,

from
sketch

</rule>

{ <rule rule>
 {
 name: —
 area: —
 }
</rule rule>





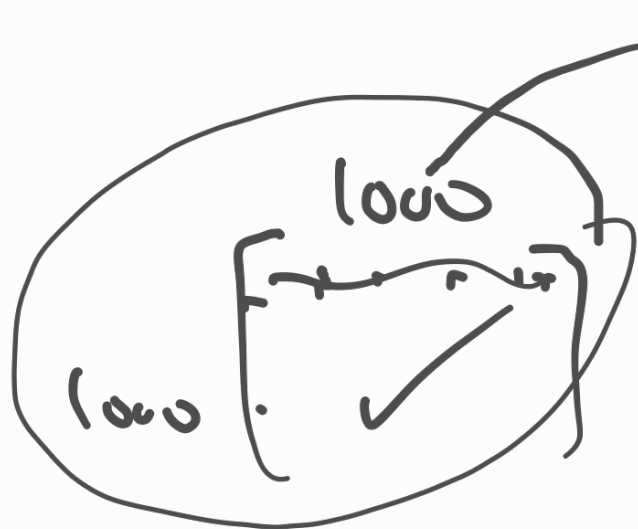
Freeze the
conv and
width

1.73

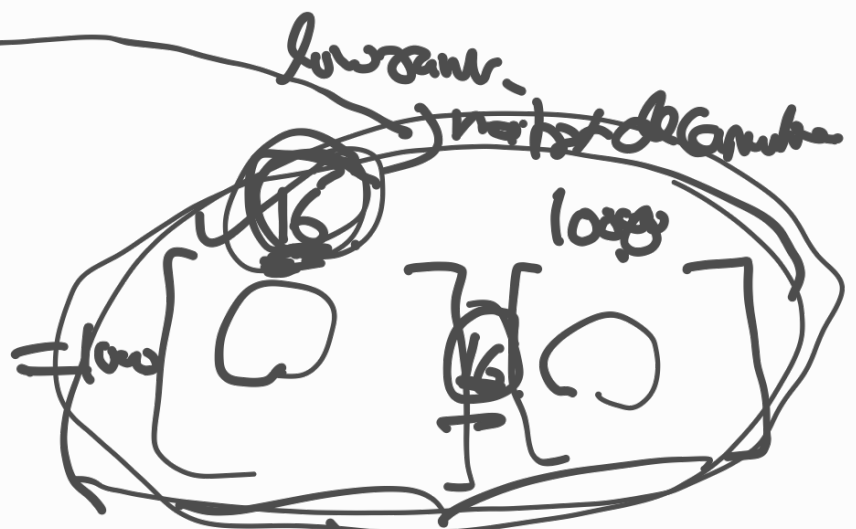
16x4
2x3



Function
with
low recommended
prod



1000x1000 param



1000x16 f

1000x16

16x16

16x16

324

16x16 param

17x17

Low Param Adapter

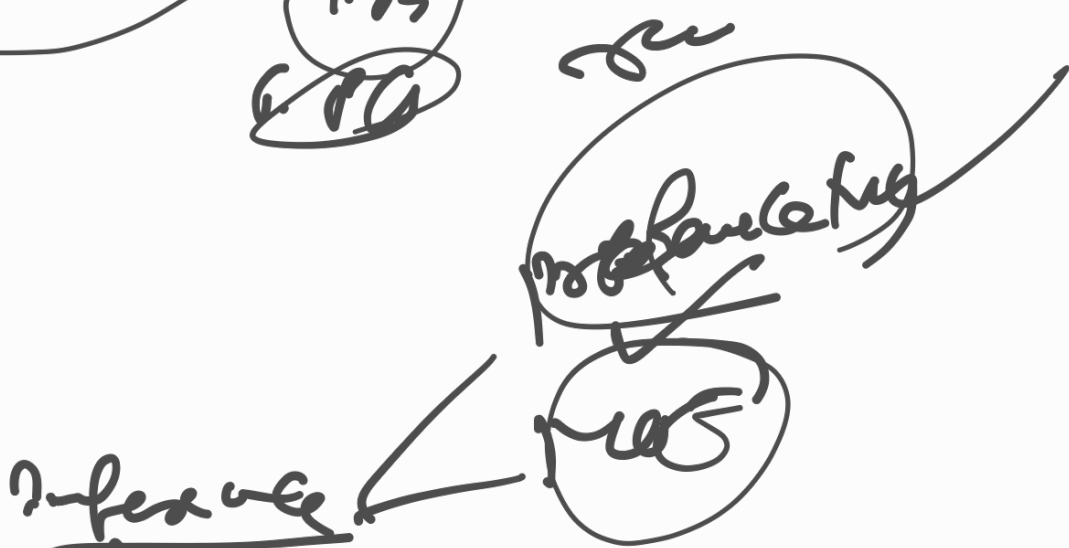
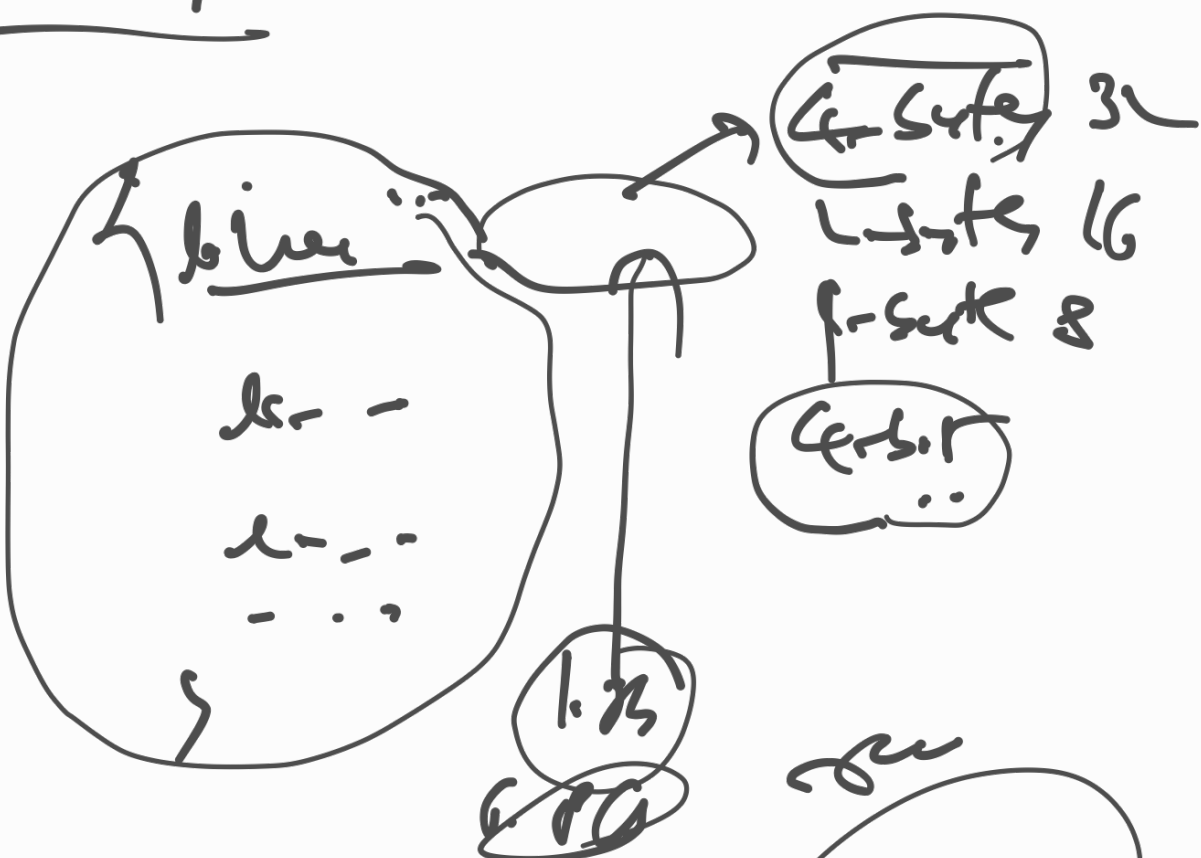
① Freeze the conv and

② create a low-param model with
adapter layers

Q loss → quantized GPT

reduce n. parameters
~~of the model~~

reduce the
x/sizd
base model
memory footprint



Inference → GPT → evaluation

Fast technique
kv-cache
speculative
decoding
batching

$$\langle a \rangle - \dots - \langle a \rangle \cdot \langle a \rangle = \langle a \rangle$$

$$\langle a \rangle \cdot \dots \cdot \langle a \rangle = \langle a \rangle$$

